Ali Muqtadir

Email address: sam24@stir.ac.uk Website: sa-muqtadir.github.io Work address: Department of Psychology, University of Stirling, FK94LA, Stirling (United Kingdom)

Skills

Human Data Collection:

• Magnetic Resonance Spectroscopy

• Magnetic Resonance Imaging

• Electroencephalography

• Transcranial Magnetic Stimulation

• Transcranial Direct Current Stimulation

• Electromyography

• Venepuncture

• Peripheral IV Cannulation

• Cognitive Testing

• Neuropsychological Assessments

• Structured Clinical Interviews

Programming:

• R

• MATLAB

Work experience

Doctoral Researcher — 01/09/2022 – Current

University of Stirling | Stirling, United Kingdom

- My PhD project focused on isolating the acute effects of football heading on cortical excitation-inhibition, neuronal structural integrity, and neural metabolites
- Used 7T MR spectroscopy to investigate heading effects on brain metabolites at the Imaging Centre for Excellence (ICE), Glasgow and Forschungszentrum Jülich
- Explored how heading changes the trajectory blood-based biomarkers of brain injury over 24 hours in a serial sampling study using ultra-sensitive immunoassays at UK DRI labs, University of Edinburgh
- Used concurrent TMS-EEG to investigate excitation-inhibition balance changes following heading exposure
- Collected and contributed heading kinematic data to computational simulations of brain biomechanics following heading impacts in collaboration with the Department of Engineering, University of Edinburgh
- Led and contributed to successful funding applications, ensured compliance with local and external ethics and governance procedures including international cooperation agreements
- Contributed to increasing technical capabilities at Stirling with the first implementation of TMS-EEG and adoption of a low-cost navigated TMS system, and at ICE, Glasgow with the first implementation of an edited MRS sequence on the 7T scanner

Research Associate — 01/08/2019 – 31/08/2022

Lahore University of Management Sciences | Lahore, Pakistan

- Worked in the areas of consumer behaviour and communication to investigate how 'viral' content spreads online and the factors that mediate the spread of misinformation or 'fake news'
- Contributed to three research projects on online misinformation, social media communication and digital content virality
- Co-authored empirical case studies applying a logistic diffusion model to real-world YouTube virality data for Bhimani et al. (2024)
- Contributed to study design and manuscript drafting for Ali et al. (2022)

Graduate Researcher — 01/02/2018 – 31/08/2018

Department of Psychosis Studies, King’s College London | London, United Kingdom

- Worked on a project investigating the effects of non-invasive brain stimulation on cognitive processes in patients with schizophrenia
- Investigated the effects of transcranial direct current stimulation (tDCS) on probabilistic learning in schizophrenia, contributing to publications in Psychiatry Research and Scientific Reports
- Administered tDCS protocols, structured clinical interviews (PANSS), and cognitive assessments within a clinical research setting

Education & Training

PhD Psychology — 01/10/2022 – Current

University of Stirling | Stirling, United Kingdom

Level in EQF: 8 | **Thesis:** The Heading Brain: Characterising the Acute Neurobiological Response to Football Heading

MSc Neuroscience — 25/09/2017 – 30/09/2018

King's College London | London, United Kingdom

Level in EQF: 7 | **Thesis:** Effect of transcranial direct current stimulation on probabilistic learning in patients with schizophrenia

BSc (Hons) Psychology 01/09/2012 – 22/12/2016

University of Central Punjab | Lahore, Pakistan

Level in EQF: 6

Publications

Social media virality: Reaching the tipping point

Bhimani, A., Zainulabdin, K., Ali, K., Ali Muqtadir, S., & Hausken, K. (2024) | 8(3), 27–41 | Journal of International Business Research and Marketing

doi.org/10.18775/jibrm.1849-8558.2015.83.3003

The effects of emotions, individual attitudes towards vaccination, and social endorsements on perceived fake news credibility and sharing motivations

Ali, K., Li, C., Zain-ul-abdin, K., & Muqtadir, S. A. (2022) | 134, 107307 | Computers in Human Behavior

doi.org/10.1016/j.chb.2022.107307

Stimulating learning: A functional MRI and behavioral investigation of the effects of transcranial direct current stimulation on stochastic learning in schizophrenia

Orlov, N. D.*, Muqtadir, S. A.*, Oroojeni, H., Averbeck, B., Rothwell, J., & Shergill, S. S. (2022) | 317, 114908 | Psychiatry Research

doi.org/10.1016/j.psychres.2022.114908

The effect of training intensity on implicit learning rates in schizophrenia

Orlov, N. D.*, Sanderson, J.*, Muqtadir, S. A.*, Kalpakidou, A. K., Michalopoulou, P. G., Lu, J., & Shergill, S. S. (2021) | 11(1), 6511 | Scientific reports

doi.org/10.1038/s41598-021-85686-5

The effect of football heading on blood biomarkers of brain injury

Muqtadir, S. A., Aston, T., Moran, C., Ietswaart, M. (2025) | ISRCTN

Pre-registration

doi.org/10.1186/ISRCTN44241334

Impact of Football Heading on TMS-EEG Indicators of Neural Excitation and Inhibition

Muqtadir, S. (2025) | OSF

Pre-registration

https://doi.org/10.17605/OSF.IO/4AC6J

Certifications

MRI Safety Training for MHRA Category B Staff 12/07/2026

NHS Scotland TURAS Learn

Peripheral I.V Cannulation 12/03/2025

Geopace Training

Phlebotomy 21/06/2023

Geopace Training

Teaching and Supervision

University of Stirling
01/10/2022 – Current

- Co-supervised a master's project investigating EEG activity patterns and natural environment exposure
- Ran weekly sessions on EEG ERP analysis in Brain Vision Analyzer for work placement students
- Supervised undergraduate research projects and ran tutorials for research design in psychology
- Conducted lab demonstrations for master's students, including practical TMS and EEG sessions
- Facilitated interdisciplinary Crucible workshops as Faculty Officer, bringing together researchers across disciplines
- Carried out marking for undergraduate assignments across multiple modules

Lahore University of Management Sciences
01/08/2019 – 31/08/2022

- Trained undergraduate researchers in eye-tracking data collection on a Gazepoint system
- Delivered teaching on research methods for undergraduate students